

SSA2 (P10592) — Review of GO:0005886

"plasma membrane" annotation

Focus: existing_go_annotation_decision · existing-go-0005886-keep-as-non-core **Term:** plasma membrane (GO:0005886) · Evidence: IBA · Ref: GO_REF:0000033 **Current action under test:** KEEP_AS_NON_CORE **Iteration:** finalized in Iteration 3 of 3

Data-integrity note (Iteration 2, QuickGO/EBI live check): In the current GO database the **only** plasma-membrane (GO:0005886) annotation for SSA2 is **HDA from**

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(SGD). There is **no IBA annotation to plasma membrane** — the IBA (GO_REF:0000033, GO_Central) pipeline propagated only cytosol/cytoplasm/nucleus to SSA2. The seed row's evidence_type: IBA for GO:0005886 therefore appears to be a **mismatch**; the reviewed term is HDA-only. The paralog **SSA1 (P10591) carries the identical single HDA PM annotation from the same paper**, consistent with shared-peptide paralog co-detection (SSA1/SSA2 ≈97% identical). This does not change the verdict but should be corrected in the review.

Executive Judgment

Verdict: Partially supported — the KEEP_AS_NON_CORE action is justified (conservative and appropriate).

The plasma-membrane annotation is real in the databases but weakly grounded. It rests on (1) a single high-throughput proteomics detection in a *stripped* plasma-membrane fraction (

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Delom et al. 2006) recorded as HDA, and (2) an IBA phylogenetic propagation from the HSP70 family (GO_REF:0000033). Neither establishes plasma membrane as a **primary site of Ssa2 function**. SSA2 is one of the most abundant cytosolic Hsp70 chaperones and has **no transmembrane domain, no signal peptide, and no lipid anchor** (UniProt P10592), so its

appearance in a membrane pellet is best explained by abundant-cytosolic-protein carryover — the same fractions routinely capture elongation factors and enolase (

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There is genuine cell-periphery biology (histatin-5 receptor at the cell envelope,

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secreted/cell-wall and extracellular-vesicle IDA annotations), but that is **cell wall/secreted**, not integral plasma membrane.

Most important caveats: (a) The evidence would also support a defensible *remove/do-not-propagate* argument for the IBA specifically; KEEP_AS_NON_CORE is the more conservative of two reasonable choices. (b) Curators should ensure the term is never treated as core and that the richer, better-evidenced peripheral terms (cell wall GO:0009277, extracellular region/vesicle) carry the cell-surface story instead.

Evidence Matrix

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confid limitat
P 16622836	Localization, HT proteomics (HDA)	Qualifies	Ssa2 is at the PM	~90 proteins from a <i>stripped</i> PM fraction (dodecyl-maltoside, IEC/LDS-PAGE + LC-MS/MS); Ssa2 detected	S. cerevisiae stripped PM	Weak - HT stu functio follow abund protein carryo likely
QuickGO/EBI live check	Database provenance	Refutes (seed evidence_type)	GO:0005886 is supported by IBA (per seed)	No IBA to PM exists ; live PM annotation is HDA-only (P 16622836). IBA (GO_REF:0000033) covers only cytosol/ cytoplasm/ nucleus. Paralog SSA1 has identical single HDA PM annotation.	GO/SGD/ GO_Central, 2026 snapshot	Strong correct seed ro eviden
UniProt P10592	Database / sequence-feature	Refutes (core)	Ssa2 is an integral/ core PM protein	No TM, no signal peptide, no lipidation; curated location = Cytoplasm; Secreted, cell wall; PM only as HDA; no "Membrane" keyword	UniProtKB curated	Strong integra residen soluble protein
P 16544286	Methodological control	Qualifies		Yeast surface/ membrane	S. cerevisiae	

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confid limitat
			PM fractions are clean	proteomics fractions also recover elongation factors & other cytoplasmic proteins	surface proteomics	Support contam interpr
P 12761219	Direct assay + mutant phenotype	Competing	Ssa1/2p works at the cell periphery	Ssa1/2p is the cell- envelope receptor for histatin-5; co- localized on cell- wall extracts AND cytosolic fractions; Δ ssa1 Δ ssa2 lowers binding/ killing	C. albicans & S. cerevisiae	Real ce surface but cel envelo PM per
P 25853343	Direct assay	Competing	Primary site of Ssa2 function	Ssa2p = major cytoplasmic Hsp70; nuclear tRNA import, binds Nup116	S. cerevisiae	Core = cytosol nuclea chaper carrier
P 29651044 ; P 10745074	Direct assay	Competing	Ssa1/2 core molecular function	Cytosolic Hsp70 chaperone activity in protein import/folding (aconitase mito import; Ape1 vacuolar transport)	S. cerevisiae	Membr associa transie client-c

Provenance: `/tmp/SSA2_G00005886_evidence_matrix.csv`,
(computed this run).

`/tmp/SSA2_CC_landscape.csv`

GO Curation Implications

- **Lead (requires curator verification): KEEP the GO:0005886 annotation but treat it as NON-CORE** — consistent with the seed action.
- It is a **CC** term. Evidence supports *presence at* but not *function at* the plasma membrane.
- Do **not** generalize it to a core location term (cytoplasm/cytosol IDA already carry the core CC).
- **Alternative lead worth flagging:** because the sole experimental support is one stripped-membrane HT proteomics hit for a hyper-abundant soluble chaperone, a curator could reasonably down-weight or not-propagate the **IBA** (mark with NOT/annotation-extension review) while retaining the HDA as non-core. This is a stronger-scrutiny option, not a recommendation to delete outright.
- Ensure the **better-evidenced peripheral terms** are the ones telling the cell-surface story: fungal-type cell wall (GO:0009277, IDA), extracellular vesicle (GO:1903561, IDA), secreted/extracellular region.

SSA2 cellular-component landscape (from UniProt/SGD): cytoplasm (IDA, core), cytosol (IDA, core), fungal-type cell wall (IDA), extracellular vesicle (IDA), fungal-type vacuole membrane (IDA), mitochondrion (IDA), nucleus (IBA), **plasma membrane (HDA + IBA → non-core, weak)**.

Mechanistic Scope

The immediate molecular function of Ssa2 is **ATP-dependent Hsp70 chaperone activity** (nucleotide-binding domain + substrate-binding domain) acting in the **cytosol/nucleus**: de novo folding, post-translational translocation across organelle membranes, tRNA nuclear import, and assembly of client complexes. Any plasma-membrane signal is **downstream/associative**, not a direct PM activity: (i) fraction co-purification of an abundant soluble protein, and (ii) transient, client-directed membrane engagement (e.g., inserting viral replication proteins into membranes,

).

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None of these convert PM into a site of intrinsic Ssa2 molecular function.

Conflicts and Alternatives

- **Contamination / frequency bias:** Ssa2's extreme cytosolic abundance makes membrane-pellet detection the expected null result; the control literature

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confirms cytoplasmic carryover in these fractions.

- **Paralog confusion:** Ssa1/Ssa2 are ~97% identical and functionally redundant; several "cell-surface Ssa" studies (e.g., histatin) cannot resolve Ssa1 vs Ssa2 vs Ssa3/4. Cell-surface claims may be family-level, not SSA2-specific. **Confirmed:** SSA1 (P10591) carries the *identical* single HDA PM annotation from the same paper

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i.e. the proteomics signal was propagated to both paralogs and cannot distinguish them.

- **Evidence-code carry-over (data integrity):** the seed labels the reviewed annotation IBA (GO_REF:0000033), but the live GO record has **no IBA to plasma membrane**; the term is HDA-only. The "IBA is consistent" rationale in the review summary does not apply and should be revised.
- **Location mis-typing:** the strongest peripheral evidence points to **cell wall/cell envelope and secreted**, which are distinct from the plasma membrane; GO:0005886 may be a coarse stand-in for these better-supported terms.
- **IBA source strength:** the phylogenetic call inherits mammalian Hsc70 PM annotations, whose own "PM" status is itself partly surface/associative.

Knowledge Gaps

1. **Is Ssa2 genuinely at the PM vs cell wall?** Checked: UniProt features (no TM/anchor), Delom proteomics (stripped PM), histatin papers (cell wall). Matters because it decides cell-wall vs PM term. Resolve with GFP/split-GFP PM-marker co-localization or protease-shaving surface proteomics with strict PM-marker controls.
2. **Paralog specificity.** Checked: sequence identity/redundancy literature. Matters for SSA2-specific annotation. Resolve with Ssa2-only tagged strains in an *ssa1Δ ssa3Δ ssa4Δ* background.

3. **Functional consequence at the PM.** Checked: no functional PM assay exists. Matters because "non-core" hinges on absence of a PM function. Resolve with PM-targeted degron/anchor-away tests for a PM-localized phenotype.
4. **Quantitative abundance rank.** The contamination argument leans on Ssa2 being among the most abundant cytosolic proteins. Checked: attempted a programmatic PaxDb lookup (YLL024C) — the PaxDb API was **not accessible (HTTP 404)**, so no numeric rank is reported here to avoid fabrication. The abundance is nonetheless well documented in the literature (Hsp70 Ssa1/Ssa2 are among the highest-copy chaperones; UniProt lists "Direct protein sequencing"). A curator can confirm the exact rank via PaxDb/SGD abundance datasets; this would further quantify carryover risk.

Discriminating Tests

- **Quantitative co-localization** of Ssa2-GFP with a bona-fide PM marker (e.g., Pma1-mCherry) vs a cell-wall marker, in live cells — distinguishes PM from wall/cytosol.
- **Fractionation with sucrose-gradient PM enrichment + abundance normalization:** does Ssa2 enrich with PM markers beyond its bulk cytosolic level, or track with soluble contaminants?
- **Surface biotinylation / trypsin-shaving** of intact spheroplasts to test true surface exposure.
- **Cross-reference** SGD/GO annotation provenance to confirm the HDA derives solely from Delom 2006 and whether the IBA has independent experimental support in the HSP70 tree.

Curation Leads (require curator verification)

- **Action:** Retain seed action **KEEP_AS_NON_CORE** for GO:0005886. Rationale is sound.
- **Correct the review metadata:** change `evidence_type` for GO:0005886 from **IBA** → **HDA** (live GO shows HDA/

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only; no IBA to PM). Revise the review summary sentence "IBA is consistent" because no PM IBA exists. Verify against QuickGO (`geneProductId=P10592&goId=GO:0005886`).

- **Candidate reference to cite in the review:**

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— verify snippet: *"Proteins from a stripped plasma membrane fraction were solubilized with the neutral and non-denaturing detergent, the n-dodecyl beta-D-maltoside."* (documents the HT/fractionation nature of the evidence).

• **Contamination-control reference:**

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— verify snippet: *"Elongation factors, mitochondrial membrane proteins and a mannosyltransferase... were also identified in this fraction."* (supports carryover interpretation).

• **Competing/peripheral biology reference:**

P 12761219

— verify snippet: *"co-localized on immunoblots of yeast beta-mercaptoethanol cell wall extracts and cytosolic fractions."*

- **Candidate refinement:** ensure cell-surface signal is represented by GO:0009277 (fungal-type cell wall, IDA) and GO:1903561 (extracellular vesicle, IDA), so PM is not over-interpreted.
- **Optional stronger-scrutiny lead:** flag the IBA GO:0005886 for reduced propagation confidence given the single abundant-protein HT source; do not delete without curator review.
- **Suggested question for curator:** Should GO:0005886 be down-graded/annotated with an evidence note distinguishing "detected in PM fraction" from "functions at PM"?

Analyses run: NCBI eutils fetch of

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UniProt P10592 feature/localization/GO-CC parse; QuickGO/EBI live annotation query for P10592 and P10591 (Iteration 2); evidence-matrix and CC-landscape tables (saved to /tmp).

Iteration-2 QuickGO result (verbatim): SSA2 GO:0005886 = 1 annotation, evidence=HDA, ref=

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by=SGD; SSA1 GO:0005886 = 1 annotation, evidence=HDA, ref=

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by=SGD. Full SSA2 CC set (16 annotations) shows IBA/GO_REF:0000033 only on GO:0005634, GO:0005737, GO:0005829.

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